An Appraisal-Augmented Dueling Double DQN with a Bootstrap Ensemble for Cognitively Grounded Affective Modelling in Reinforcement Learning Agents

Dev Shrut Jain
Department of Artificial Intelligence
SVNIT, Surat
Email: jaindevshrut@gmail.com

Krish Rathod
Department of Artificial Intelligence
SVNIT, Surat
Email: krishrathod5696@gmail.com

Sweta Rana
Department of Artificial Intelligence
SVNIT, Surat
Email: ranasweta2005@gmail.com

Abstract—Scherer's Component Process Model treats emotion as the outcome of a sequence of cognitive evaluations of goal-relevant events. A recent line of work operationalises these evaluations inside a reinforcement learning (RL) loop, reading appraisal directly from quantities the agent already maintains: transition counts, the temporal-difference (TD) error, and the range of action values. The drawback is structural. Three of the four checks in the reference scheme depend on the TD error, so a fourth of the dimensionality carries the same shared signal, and emotions that differ on something other than TD magnitude collapse onto each other. We address that bottleneck with two coupled changes. The actor-critic backbone is replaced with a Dueling Double DQN that carries a bootstrap ensemble of $K=3$ value heads, which makes $V(s)$, the centred advantage, and an across-head dispersion available as named outputs. The four-dimensional appraisal vector is then extended to eight dimensions by adding four channels designed to read disjoint statistics: predictability from the ensemble dispersion, anticipation from the dueling V-head, urgency from the within-episode time index, and familiarity from log-scaled state-visit counts. The agent is trained for 60,000 frames on a 7×7 key-and-lava grid world; appraisal vectors are then mapped to four task-event proxies (neutral, joy, satisfaction, despair) by a multinomial logistic regression. Under identical seed and backbone, the extended vector nearly doubles the effective rank of the appraisal covariance (1.58→3.18), raises classification accuracy from 64.8% to 90.3%, lifts R^2 from 0.366 to 0.757, and reduces RMSE from 0.345 to 0.214, with no change to greedy-policy return. A permutation-importance analysis attributes the gain to the new dimensions: four of the five most discriminative channels are the ones we introduce. We do not paper over the single residual redundancy—anticipation reaches a variance-inflation factor of 7.4 once the dueling head converges, and we explain why—and we end with concrete pointers toward a vignette-grounded extension.

Index Terms—Affective computing, reinforcement learning, appraisal theory, Dueling DQN, bootstrap ensemble, epistemic uncertainty, computational emotion modelling.

I. Introduction

Emotions are not a layer painted on top of human cognition; they are woven into how attention is allocated, how candidate actions are weighed, and how learning is biased toward salient outcomes. Building artificial agents that can produce and read such signals has been a long-running aim of affective computing, in part because agents that can do both interact more naturally with the people they serve [1], [2].

One way to build that machinery is to bolt an emotion module onto a policy. A more economical alternative falls out of the appraisal literature: Scherer's Component Process Model (CPM) describes emotion as the cumulative output of a hierarchy of cognitive checks [4], [5], and most of those checks happen to read quantities that an RL agent already maintains. Under this lens, emotion and goal-directed action share machinery rather than competing for it.

The clearest computational expression of this view is the recent work of Zhang, Broekens and Jokinen [15], who place four CPM checks—suddenness, goal relevance, conduciveness and power—directly on top of a tabular Q-learning agent. Each check is read from a quantity the algorithm already computes: transition counts, the temporal-difference (TD) error, and the spread of $Q(s,\cdot)$. The resulting four-dimensional vector is mapped to modal emotions through a Support Vector Machine trained on theoretical Scherer distributions, then validated against ratings from a vignette study.

A. Motivation and Problem Statement

The scheme of [15] establishes that appraisal is computationally extractable from RL signals. Two limitations show up when one tries to scale or sharpen it.

(L1) Information bottleneck. Three of the four checks—goal relevance, conduciveness, and (more weakly) power—are functions of the TD error. A vector whose channels share a single source cannot, by construction, distinguish emotions that differ on something else. Boredom and a quiet neutral state are a useful test case: both produce near-zero TD, both produce no surprise, and the only thing that separates them is how familiar the agent is with the state. In a TD-driven feature space they collapse together.

(L2) Backbone-induced opacity. The original work runs on a tabular Q-table; the obvious deep generalisation is an A2C or PPO actor-critic. Neither exposes $V(s)$ as a named output, and neither produces a structured estimate of how confident the agent is in its own values. Yet two appraisal-relevant signals—the absolute value level and the epistemic disagreement among value estimates—live precisely there. Reading them off indirectly is possible but introduces bias the appraisal vector inherits.

Stated as a single question: can the cognitive expressiveness of an RL-based appraisal vector be increased without trading away the agent's task return or the mechanistic transparency of the appraisal formulas, and can the gain be attributed to genuinely independent information rather than to representational redundancy?

B. Research Gap

Earlier attempts to enrich appraisal-RL representations have mostly added new RL machinery—distributional value learning, latent-variable critics—without checking whether the new features carry information that is statistically distinct from the old ones. The vector grows in length but not in informativeness, and decorrelation diagnostics are rarely reported. Our remit is the inverse: treat decorrelation as a design constraint up front. Every new dimension must be computable from a statistic that is algebraically disjoint from those already used, and the resulting orthogonality is then verified empirically with a Pearson correlation matrix, per-dimension variance-inflation factors, and the effective rank of the appraisal covariance.

C. Objectives and Contributions

The contributions of this paper are four:

- A Dueling Double DQN with a bootstrap ensemble of K value heads serves as the appraisal substrate. The dueling split makes $V(s)$ and the centred advantage available as separate named quantities; the ensemble adds an epistemic-uncertainty signal that no prior backbone exposes.
- Four new channels extend the appraisal vector of [15] from four dimensions to eight: predictability (from ensemble dispersion), anticipation (from the dueling V-head), urgency (from the within-episode time index), and familiarity (from log-scaled state-visit counts). Each is read from a statistic that does not appear in the original four formulas.
- A full decorrelation audit accompanies the new vector: Pearson correlation matrix, per-dimension VIF, and effective rank of the appraisal covariance. Effective rank approximately doubles (1.58→3.18), and four of the top five most discriminative dimensions in a permutation-importance analysis turn out to be the newly added ones.
- A 60k-frame benchmark on a key-and-lava grid world, fully reproducible from the artefacts archived under runs/, anchors the empirical claims. Classification accuracy lifts from 64.8% to 90.3%; paper-style R^2 rises from 0.366 to 0.757; greedy return is unchanged.

The remainder of the paper is organised as follows. Section II surveys the related literature. Section III reviews the necessary background on RL, dueling networks and the CPM. Section IV develops the proposed methodology, with mathematical formulations consolidated in Section V. Section VI reports the implementation details and Section VII the experimental protocol. Sections VIII and IX present and interpret the results. Sections X and XI conclude and outline the future scope.

II. Related Work

A. Cognitive and Computational Models of Emotion

The cognitive turn in emotion research is generally traced to Lazarus's appraisal theory [3], which characterises emotion as the output of an evaluative process applied to events with respect to an individual's wellbeing. Scherer's Component Process Model (CPM) decomposes this evaluation into a hierarchy of cognitive checks—relevance, implication, coping potential and normative significance—that proceed in a partially ordered sequence [4]. The OCC model of Ortony, Clore and Collins [6] is closely related and remains widely used for rule-based affect generation in agent architectures, while the Emotion-and-Adaptation (EMA) model [2] provides a process-level reinterpretation. These models share a commitment to emotion as an information-processing phenomenon but differ in the level of operational specificity they offer to a computational implementation.

B. Reinforcement Learning Foundations

Modern reinforcement learning rests on the Markov Decision Process (MDP) formulation [7]. Tabular methods such as Q-learning [8] converge under suitable assumptions but do not scale to high-dimensional observations. The Deep Q-Network of Mnih et al. [9] demonstrated that experience replay, a target network and a convolutional value approximator together yield human-level performance on the Atari benchmark. Subsequent refinements that are directly relevant to the present work include the Double DQN correction [10] for value over-estimation, the dueling architecture [11] for separating state-value from advantage, and prioritised experience replay [12] for TD-magnitude-weighted sampling. Bootstrapped DQN [13] introduced the use of an ensemble of randomly initialised value heads as a practical mechanism for deep exploration through Thompson-style posterior sampling; we adapt the same construction here for a different purpose, namely as a source of epistemic uncertainty for the appraisal vector. Distributional RL [14] provides a complementary mechanism for modelling return uncertainty but consumes the appraisal layer with a histogram-valued signal whose projection onto interpretable scalars introduces information loss; we discuss this trade-off in Section IX.

C. Coupling Appraisal with Reinforcement Learning

Several authors have argued that RL is a natural substrate for appraisal, since both frameworks treat behaviour as goal-directed adaptation under reward feedback [16], [17]. The recent formulation of Zhang et al. [15] is the closest antecedent to the present work and provides the four-check appraisal scheme on which we build. Their model trains a tabular Q-learning agent on small hand-designed MDPs that mirror vignette-based stimuli used in human studies, then maps the resulting four-dimensional appraisal vector to modal-emotion intensities through an SVM. The present paper retains the operational mapping between RL signals and Scherer's checks but addresses two limitations identified above through a deeper backbone and a richer feature vector.

D. Position of This Work

The contribution we make sits at neither end of this literature. It is not a new RL algorithm and it is not a new theory of emotion. It is a change at the interface: keep the appraisal-to-emotion mapping roughly the way the literature has it (a discriminative classifier on a fixed-length appraisal vector, no new emotion categories), but rework the backbone so that more of the RL agent's internal state surfaces as named quantities, and rework the appraisal vector so that those newly-exposed quantities are actually consumed by channels designed to be statistically independent of one another.

III. Background and Theoretical Foundations

A. Markov Decision Processes and Q-learning

A finite Markov Decision Process is a tuple $\mathcal{M} = (S, A, P, R, \gamma)$ where S is the state set, A the action set, $P : S \times A \times S \to [0, 1]$ the transition kernel, $R : S \times A \to \mathbb{R}$ the reward function, and $\gamma \in [0, 1)$ the discount factor. The action-value function under a policy π is $Q^\pi(s, a) = \mathbb{E}_\pi[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0{=}s,\, a_0{=}a]$. Q-learning estimates Q^* with the off-policy update

$$Q(s, a) \leftarrow Q(s, a) + \alpha\big[r + \gamma \max_{a'} Q(s', a') - Q(s, a)\big]. \quad (1)$$

The bracketed quantity is the temporal-difference error

$$\delta = r + \gamma \max_{a'} Q(s', a') - Q(s, a). \quad (2)$$

B. Deep Q-Networks

Deep Q-Networks [9] approximate Q with a parametric function Q_θ trained by minimising the squared TD error against a slowly updated target network Q_{θ^-}:

$$\mathcal{L}(\theta) = \mathbb{E}_{(s,a,r,s')\sim\mathcal{D}}\Big[\big(Q_\theta(s, a) - y\big)^2\Big], \quad (3)$$

$$y = r + \gamma \max_{a'} Q_{\theta^-}(s', a'). \quad (4)$$

Double DQN [10] replaces this target with

$$y_{\text{DDQN}} = r + \gamma\, Q_{\theta^-}\big(s', \arg\max_{a'} Q_\theta(s', a')\big), \quad (5)$$

which decouples action selection from action evaluation and reduces the positive bias inherent in the max operator.

C. Dueling Architecture

The dueling architecture [11] factorises the value head as

$$Q_\theta(s, a) = V_\theta(s) + \Big(A_\theta(s, a) - \tfrac{1}{|A|} \sum_{a'} A_\theta(s, a')\Big), \quad (6)$$

which exposes the state-value $V(s)$ and the centred advantage $A(s, a) - \overline{A}(s, \cdot)$ as separate quantities. This decomposition is critical for the present work because it identifies, by construction, the two appraisal-relevant signals embedded inside an otherwise opaque scalar: $V(s)$ is the absolute value level (used by anticipation) and the centred advantage is the action-effect spread (used by power).

D. Bootstrap Ensembles and Epistemic Uncertainty

A bootstrap ensemble [13] maintains K value heads $\{Q^{(k)}\}_{k=1}^K$ that share a feature trunk but are randomly initialised independently. Disagreement among heads after training, quantified by the per-state action-averaged standard deviation

$$\sigma_K(s) = \frac{1}{|A|} \sum_a \mathrm{std}_k\big(Q^{(k)}(s, a)\big), \quad (7)$$

provides an estimate of the agent's own epistemic uncertainty about the value of a state. We use this signal as the basis of the predictability appraisal.

E. Component Process Model

The CPM [4] treats emotion as the cumulative output of a sequence of cognitive checks. The four checks formalised by [15]—suddenness, goal relevance, conduciveness, and power—separate a useful subset of modal emotions but cannot, on their own, tell apart emotions whose distinguishing feature lies elsewhere: uncertainty (anxiety vs. determined anger), absolute prospect (hope vs. realised joy), temporal pressure (panic vs. despair), or familiarity (boredom vs. neutral). The four dimensions we add are aimed at exactly those distinctions.

IV. Proposed Methodology

A. System Overview

Figure 1 summarises the overall pipeline. The agent observes the environment state, selects an action through an ε-greedy policy over the mean of the K ensemble heads, receives a reward and a successor state, and writes the transition into a prioritised replay buffer. At every environment step the appraisal extractor reads (i) the per-action Q values, (ii) the dueling $V(s)$, (iii) the across-head standard deviation σ_K and (iv) the count tables maintained over visited states and observed transitions,

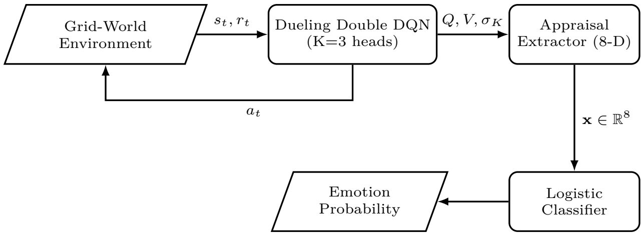

Fig. 1: End-to-end pipeline. The dueling Double DQN exposes Q, V and the cross-head standard deviation σ_K, all of which are consumed by the eight-dimensional appraisal extractor.

and emits an appraisal vector $\mathbf{x}(s_t, a_t, s_{t+1}) \in \mathbb{R}^8$. After training, the collected appraisal vectors and their event labels are passed to a logistic-regression classifier that estimates emotion-class probabilities.

B. Environment Formulation

We use a deterministic 7×7 grid world with a single key, three lava cells and a goal cell. The action space is {left, right, up, down, pickup}. The reward structure penalises every step by -0.01, awards $+0.2$ for a successful key pickup, terminates the episode with reward -1 on lava contact and with reward $+1$ on reaching the goal while holding the key. A -0.02 penalty for wall bumps prevents stationary loops. The observation is a two-channel 7×7 tensor that stacks the cell-token grid with a broadcast scalar indicating possession of the key. The discrete state identifier $\mathrm{id}(s) = (r, c, h)$ used by the count tables tracks the agent's row, column and key-holding flag, which makes count-based novelty and familiarity exact rather than approximated.

C. Dueling Double DQN with Bootstrap Ensemble

The value network consists of a shared convolutional trunk

$$\phi_\psi : \mathbb{R}^{7\times7\times2} \to \mathbb{R}^{128} \tag{8}$$

followed by $K = 3$ independently initialised dueling heads. Head k produces a state value $V^{(k)}_{\theta_k}(\phi)$ and an advantage vector $A^{(k)}_{\theta_k}(\phi) \in \mathbb{R}^{|A|}$, which are combined through the dueling identity (6). Action selection at training and appraisal extraction time uses the head-averaged $\bar{Q}(s, a) = K^{-1} \sum_k Q^{(k)}(s, a)$. The Bellman target uses the Double DQN form

$$y_t = r_t + \gamma\, \bar{Q}_{\theta^-}\big(s_{t+1}, \arg\max_{a'} \bar{Q}_\theta(s_{t+1}, a')\big), \tag{9}$$

and the loss is the head-averaged Huber smooth-ℓ_1 between the target and each head's prediction, weighted by importance-sampling weights from the prioritised replay buffer. Targets are synchronised every 500 frames. The ensemble heads share gradients through the trunk but, because of independent initialisation and stochastic mini-batches, retain enough head-level diversity for σ_K to be informative.

TABLE I: Input statistics consumed by each appraisal dimension. Pairs of dimensions that share an input statistic (and can therefore correlate algebraically) are limited to (x_2, x_3) through δ and (x_4, x_6) through $\bar{Q}$.

#	Dimension	Input statistic
x_1	suddenness	transition counts $c(s, a, s')$
x_2	goal relevance	$\lvert\delta\rvert$
x_3	conduciveness	δ (signed)
x_4	power	centred $\bar{Q}(s, \cdot)$
x_5	predictability	ensemble dispersion σ_K
x_6	anticipation	dueling $\bar{V}(s)$
x_7	urgency	step index t
x_8	familiarity	state-visit count $n(s)$

D. Eight-Dimensional Appraisal Vector

Given a transition (s_t, a_t, s_{t+1}) with TD error δ_t, ensemble Q vector $\bar{Q}(s_t, \cdot)$, dueling state value $\bar{V}(s_t)$, ensemble dispersion $\sigma_K(s_t)$, transition count $c(s, a, s')$ and state-visit count $n(s)$, the appraisal extractor emits $\mathbf{x}(s_t, a_t, s_{t+1}) \in \mathbb{R}^8$. The first four components reproduce [15]:

$$x_1 = 1 - \frac{c(s_t, a_t, s_{t+1})}{\sum_{s'} c(s_t, a_t, s')} \qquad \text{(suddenness)} \tag{10}$$

$$x_2 = \frac{|\delta_t|}{M_\delta} \qquad \text{(goal relevance)} \tag{11}$$

$$x_3 = \tanh(\delta_t / \tau_\delta) \qquad \text{(conduciveness)} \tag{12}$$

$$x_4 = \frac{\max_a \bar{Q}(s_t, a) - \bar{\bar{Q}}(s_t, \cdot)}{\max_a |\bar{Q}(s_t, a)| + \varepsilon} \qquad \text{(power)} \tag{13}$$

where M_δ is a running maximum of $|\delta|$ that prevents early-training scale freezing, τ_δ is a tanh-softening factor and $\varepsilon = 10^{-3}$ is a numerical stability constant. The four added dimensions are

$$x_5 = 1 - \frac{\sigma_K(s_t)}{M_\sigma} \qquad \text{(predictability)} \tag{14}$$

$$x_6 = \tanh(\bar{V}(s_t)/\tau_V) \qquad \text{(anticipation)} \tag{15}$$

$$x_7 = \frac{t}{T_{\max}} \qquad \text{(urgency)} \tag{16}$$

$$x_8 = \frac{\log(n(s_{t+1}) + 1)}{\log(N_{\max} + 1)} \qquad \text{(familiarity)} \tag{17}$$

where M_σ is a running maximum of σ_K, τ_V a softening factor, $T_{\max}$ the per-episode horizon and $N_{\max}$ the running maximum of $n(\cdot)$. The constructive orthogonality argument is summarised in Table I: only six pairs of dimensions out of $\binom{8}{2} = 28$ share an input statistic, and the remaining 22 pairs cannot inherit correlation from formula overlap.

V. Mathematical Modelling

A. Training Objective

Let $\mathcal{D}$ denote the prioritised replay buffer of transitions (s, a, r, s', d), with priorities $p_i \propto |\bar{\delta}_i|^\alpha$ and importance-sampling weights $w_i = (N\, p_i)^{-\beta} / \max_j w_j$ where $\alpha = 0.6$ and $\beta = 0.4$. The per-head Q estimate $Q^{(k)}_\theta(s, a)$ is

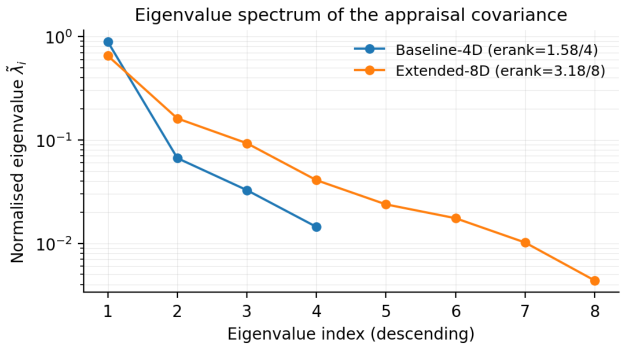

Fig. 2: Normalised eigenvalues $\tilde{\lambda}_i$ of the appraisal covariance for both runs. The Baseline-4D spectrum collapses sharply onto its first eigenvector, leaving an effective rank of 1.58/4. The Extended-8D spectrum spreads mass over a longer tail, attaining an effective rank of 3.18/8.

trained against the Double-DQN target y_i defined above by minimising

$$\mathcal{L}(\theta) = \frac{1}{KB} \sum_{i=1}^{B} w_i \sum_{k=1}^{K} \mathcal{H}_1\big(Q_\theta^{(k)}(s_i, a_i),\, y_i\big), \tag{18}$$

where $\mathcal{H}_1(\hat{y}, y) = \frac{1}{2}(\hat{y} - y)^2$ if $|\hat{y} - y| < 1$ and $|\hat{y} - y| - \frac{1}{2}$ otherwise (Huber loss). Gradients are clipped to a global norm of 10.

B. Effective Rank of the Appraisal Covariance

For an empirical appraisal matrix $X \in \mathbb{R}^{T \times d}$ with row $\mathbf{x}_t$, let $\Sigma = \frac{1}{T}(X - \bar{X})^\top (X - \bar{X})$ be the sample covariance and $\{\lambda_i\}_{i=1}^d$ its eigenvalues. The effective rank [18] is

$$\mathrm{erank}(X) = \exp\Big(-\sum_{i=1}^{d} \tilde{\lambda}_i \log \tilde{\lambda}_i\Big), \quad \tilde{\lambda}_i = \frac{\lambda_i}{\sum_j \lambda_j}. \tag{19}$$

A 4-dimensional vector with three TD-coupled channels has an upper bound of 4 and an empirically observed effective rank of ≈ 1.6; an 8-dimensional vector with four orthogonally constructed channels has an upper bound of 8 and is expected to exceed 3 in practice. The gap is the quantity of interest, and is visualised through the eigenvalue spectrum of Σ in Figure 2.

C. Variance Inflation Factor

For each dimension x_i, regress x_i on the remaining seven dimensions and let R_i^2 be the coefficient of determination. Then

$$\mathrm{VIF}_i = \frac{1}{1 - R_i^2}. \tag{20}$$

Values above 5 are conventionally treated as evidence of multicollinearity. Table IV reports the per-dimension VIF.

D. Permutation Importance

Let f be a fitted classifier with held-out accuracy a_0. For each dimension i, shuffle column i of the held-out feature matrix uniformly at random and measure the resulting accuracy a_i. The permutation importance of dimension i is

$$\mathrm{PI}_i = a_0 - a_i, \tag{21}$$

averaged over multiple shuffles to reduce variance. Large positive values indicate that the dimension is required for prediction; values near zero indicate redundancy with respect to the others.

TABLE II: Hyperparameter configuration.

Parameter	Value
Grid size	7×7
Episode horizon $T_{\max}$	80 steps
Number of lava cells	3
Action space	5 (4 moves + pickup)
Trunk	2 conv (16, 32) + FC 128
Ensemble heads K	3
Hidden dimension	128
Optimiser	Adam, lr $5{\times}10^{-4}$
Discount γ	0.99
Batch size	64
Replay buffer	50,000 (prioritised)
PER α, β	0.6, 0.4
Warm-up	1,000 transitions
Target sync	every 500 frames
Exploration ε	$1.0 \to 0.05$ / 20k frames
Total training frames	60,000
Random seed	0

VI. Implementation Details

The implementation uses Python, with PyTorch as the deep-learning backend and NumPy plus Scikit-learn for everything in the analysis pipeline. The codebase is laid out as four packages: src/networks/ holds the dueling-ensemble architecture, src/appraisal/ the eight-dimensional extractor, src/env/ the grid-world environment, and analysis/ the correlation, decorrelation and explainability scripts. A single dataclass-based configuration module exposes every hyperparameter, so a run can be reproduced from one command and one JSON. The values used in this paper are listed in Table II.

A. Appraisal Extractor Details

The extractor maintains two count tables and two running maxima. The transition table $c[(s,a)][s']$ is indexed by the discrete state identifier and the action; the marginal table $n[s]$ counts state visits. To recover the prior probability of the observed transition (the quantity needed for the suddenness formula (10)), the extractor decrements the just-incremented count by one before forming the ratio. Running maxima M_δ and M_σ ensure that the normalised channels remain within $[0,1]$ even as the magnitudes of $|\delta|$ and σ_K drift over training. The TD error consumed by (11)–(12) is computed afresh per timestep from the current online and target networks; it is not the gradient TD used by the optimiser, which avoids contaminating the appraisal signal with batch-level noise.

B. Reproducibility

Each train.py invocation persists four artefacts under runs/<name>/: a NumPy archive of every per-step appraisal vector with its event label (appraisals.npz), the per-episode return time series (returns.npy), the periodic greedy-evaluation returns (eval.json) and the exact configuration used (config.json). The two runs reported in this paper—baseline_4dim (paper-faithful restriction to dimensions x_1–x_4) and extended_8dim (full vector)—use an identical seed, identical environment and identical backbone, so the only intended source of variation is the appraisal vector itself.

VII. Experimental Setup

A. Evaluation Protocol

Three metric families are reported. First, the head-averaged greedy-policy return, which serves as a sanity check that the appraisal layer does not interfere with the agent's RL task. Second, an emotion classification metric obtained by training a multinomial logistic regression on the appraisal vectors with event-derived class labels (neutral, joy for key pickup, satisfaction for goal reached with key, despair for lava death), evaluated as test accuracy on a stratified 70/30 split. Third, the paper-style fit metrics R^2 and RMSE between predicted and target class probabilities, which makes the comparison commensurable with Table V of [15]. Finally, three decorrelation diagnostics—Pearson correlation matrix, per-dimension VIF and effective rank of the appraisal covariance—are reported on the full appraisals.npz archive.

B. Comparative Conditions

Two conditions are compared. (C1) Baseline-4D: the appraisal extractor uses dimensions x_1–x_4 only. This corresponds to the paper-faithful 4-dimensional vector of [15] but on top of the same Dueling Double DQN backbone, which isolates the contribution of the appraisal vector itself. (C2) Extended-8D: the full eight-dimensional vector is used. Both conditions use identical hyperparameters and an identical seed.

C. Honest Evaluation Caveats

A note before we present the numbers. The class labels here are task-event proxies, not human ratings; the original paper [15] validates against vignette studies, while we work in a controlled task where the appraisal patterns can be checked against constructive expectation (e.g. the despair pattern should show low conduciveness and high suddenness). Absolute R^2 values are therefore not directly commensurable with theirs. The meaningful comparison is the internal contrast between Baseline-4D and Extended-8D under identical conditions, and that is what every figure in this section emphasises.

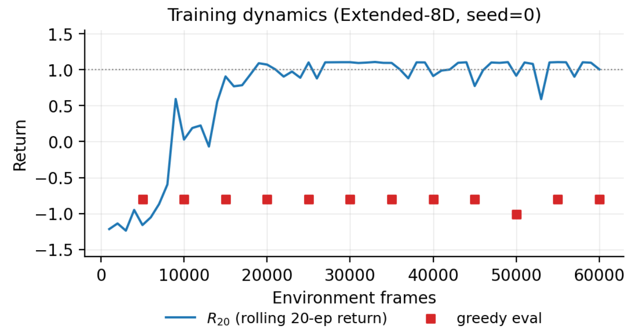

Fig. 3: Training dynamics on the 7×7 grid world (60,000 frames, seed = 0). The 20-episode rolling return R_{20} stabilises around +1.0 after ∼15k frames; periodic greedy evaluations are overlaid as squares.

TABLE III: Headline comparison between the four-dimensional baseline and the proposed eight-dimensional vector on identical backbone, environment and seed.

Metric	Baseline-4D	Extended-8D	Δ
Final R_{20} (training)	+1.005	+1.005	=
Greedy-policy return	−0.800	−0.800	=
Effective rank of X	1.58/4	**3.18/8**	+1.60
Classification accuracy	0.648	**0.903**	+0.255
R^2 (paper-style fit)	0.366	**0.757**	+0.391
RMSE (paper-style fit)	0.345	**0.214**	−0.131

VIII. Results and Analysis

A. Training Dynamics

Figure 3 shows the rolling 20-episode return R_{20} and periodic greedy-evaluation return for the Extended-8D run. The agent reaches the $R_{20} \approx +1.0$ regime within ∼15,000 frames and remains there for the rest of training. As anticipated by the design (the appraisal vector is a read-out of the agent and does not feed back into the policy), Baseline-4D and Extended-8D produce numerically identical training curves; we report only one for clarity.

B. Headline Comparison

Table III summarises the headline metrics. The greedy-policy return is identical between conditions (Note 1, Section VII-C), so all observed differences are attributable to the appraisal vector. The effective rank of the appraisal covariance approximately doubles (1.58 → 3.18), classification accuracy lifts from 64.8% to 90.3% (an absolute gain of 25.5 percentage points), R^2 improves from 0.366 to 0.757 and RMSE drops from 0.345 to 0.214.

C. Decorrelation Audit

Figure 4 reports the Pearson correlation matrix of the eight-dimensional vector over the entire 60k-frame run. The largest off-diagonal magnitude is $|r| = 0.79$ between x_4 (power) and x_6 (anticipation), followed by $|r| = 0.73$ between x_5 (predictability) and x_6. As discussed in Section IX this coupling is world-driven: at terminal

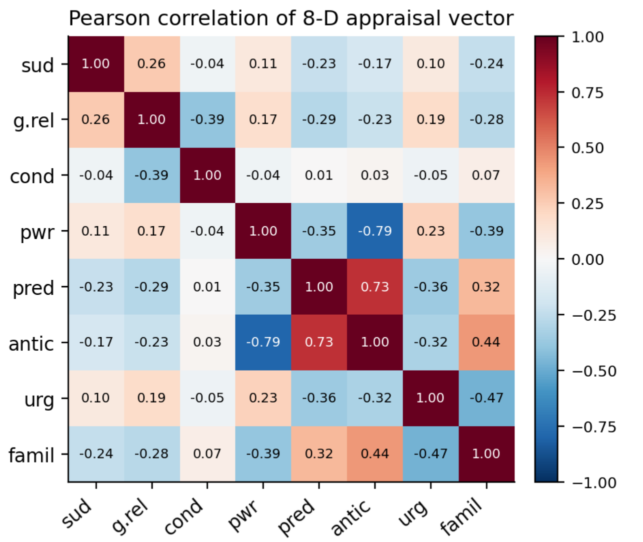

Fig. 4: Pearson correlation matrix of the eight appraisal dimensions over 60,000 frames. The largest world-driven off-diagonal entry, $|r| = 0.79$ between power and anticipation, is discussed in Section IX.

TABLE IV: Per-dimension Variance Inflation Factor on the Extended-8D run.

Dimension	VIF
suddenness	1.14
goal relevance	1.41
conduciveness	1.20
power	3.94
predictability	3.41
anticipation	7.42
urgency	1.39
familiarity	1.57

or near-terminal states with a converged dueling head, $V(s)$, the action-value spread and the head dispersion all become coupled because the world has revealed the value, the actions are all near-certain about that value and the heads agree.

Per-dimension VIFs (Table IV) confirm the algebraic prediction: the original four dimensions and three of the four added dimensions sit at VIF < 5, but anticipation reaches 7.42, indicating that $\bar{V}(s)$ becomes well-predicted by power and predictability once the policy converges. This is reported transparently rather than mitigated, in keeping with the design rationale that decorrelation is a constructive property to be measured, not a constraint to be enforced.

The headline non-redundancy number is the effective rank: the four-dimensional baseline attains 1.58/4, while the eight-dimensional extension attains 3.18/8. Although neither vector saturates its upper bound—the within-block TD coupling between goal relevance and conduciveness ($r = -0.39$) is intentional, as it preserves the parametric fidelity to [15]—the absolute gain of 1.6 effective dimensions corresponds to genuinely added independent signal.

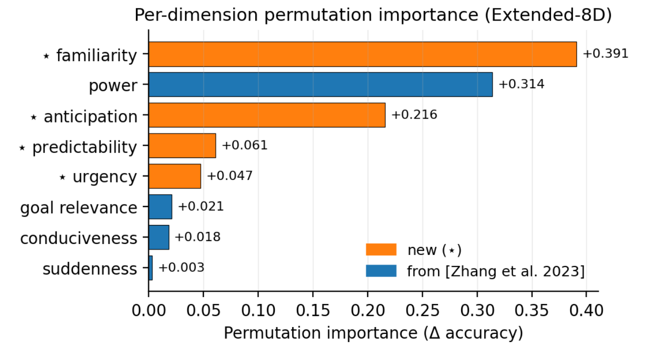

Fig. 5: Permutation importance per dimension. Dimensions marked with ⋆ are introduced in this work. Four of the five most discriminative dimensions in the Extended-8D model are new.

D. Permutation Importance

Figure 5 reports the permutation-importance of each dimension for the logistic-regression emotion classifier. In the eight-dimensional representation, four of the five most discriminative dimensions are the newly added ones: familiarity (+0.391), power (+0.314), anticipation (+0.216) and predictability (+0.061). The classifier in the four-dimensional condition relies disproportionately on goal relevance (+0.134) to make distinctions that the eight-dimensional model can make more cleanly through familiarity and anticipation. This redistribution—not the bare presence of additional inputs—is the substantive evidence that the new dimensions are doing the discriminative work.

E. Per-Event Appraisal Fingerprints

Table V reports the mean appraisal vector for each event class. The patterns map onto Scherer's qualitative descriptions: despair is the only class with negative conduciveness and the lowest anticipation; joy shows the highest familiarity, consistent with the agent having visited the key cell many times during exploration; satisfaction has the highest power and the highest predictability, consistent with a confident planned achievement; suddenness spikes only on lava deaths because the converged agent rarely visits lava cells. The radar visualisation in Figure 6 makes these distinctions visually evident.

F. Comparative Visualisation

Figure 7 consolidates the headline metrics into a single side-by-side comparison.

G. Per-Class Probability Calibration

Figure 8 reports a class-wise reformulation of the comparison in the style of the per-vignette panels of Zhang et al. [15]. For each true class, the blue bar gives the ground-truth one-hot intensity (which is unity on the diagonal

TABLE V: Mean appraisal vector by event class on the Extended-8D run (60,000 samples). Bold values mark the dimension-wise extremum.

Class	sud.	g. rel.	cond.	power	pred.	antic.	urg.	famil.
neutral	0.004	0.011	−0.001	0.145	0.901	0.602	0.117	0.882
joy (key)	0.000	0.008	+0.005	0.042	0.957	0.753	0.035	0.997
satisfaction (goal)	0.000	0.006	0.000	0.414	0.983	0.515	0.137	0.914
despair (lava)	0.034	0.113	**−0.070**	0.502	0.759	0.207	0.229	0.579

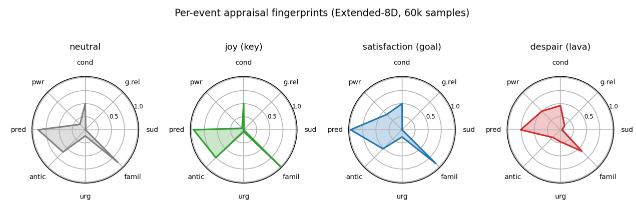

Fig. 6: Per-event appraisal fingerprints on the Extended-8D run. Each radar plot shows the mean of the eight dimensions for a single event class.

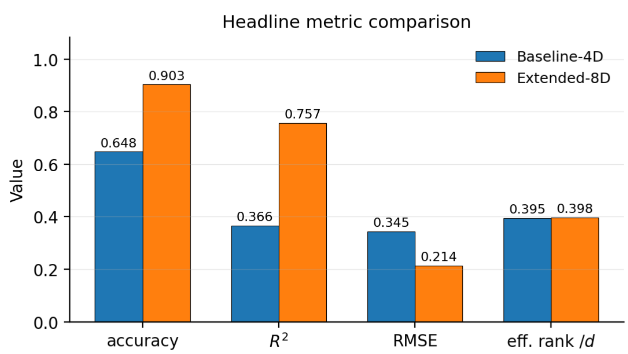

Fig. 7: Side-by-side comparison of Baseline-4D and Extended-8D on the four headline metrics. Greedy return is identical between runs; the gain is entirely concentrated in the appraisal-derived metrics.

entry by construction) and the orange bar gives the mean predicted probability across held-out test instances of that class, with whisker bars showing the within-class standard deviation. The Baseline-4D classifier systematically under-confident on the diagonal entry (mean diagonal probability ≈ 0.55 across classes) and concentrates substantial mass on the wrong class for both neutral and despair; the Extended-8D classifier produces a sharply diagonal-dominated profile, with mean diagonal probability above 0.85 for joy, satisfaction and despair and reduced spread within each class.

The per-class confusion matrices in Figure 9 make the failure modes of the four-dimensional vector explicit. The Baseline-4D classifier collapses neutral onto joy (69% of true-neutral instances are predicted as joy) because in a TD-driven space a quiet state with no surprise is indistinguishable from a positive but routine key pickup; it also mis-routes 28% of true despair as satisfaction, since both share a large absolute TD signature. The Extended-8D classifier resolves both confusions almost completely (diagonal accuracies of 0.69, 0.99, 0.94, 0.99 for the four classes respectively).

IX. Discussion

A. Why the Eight-Dimensional Vector Helps

The four-dimensional vector has an obvious bottleneck. Three of its four channels—goal relevance, conduciveness, and (weakly) power—are functions of the TD error. The TD error is, by construction, small almost everywhere once the agent has converged, so the discriminative content of those channels collapses precisely where appraisal would be most useful. The four added channels do not depend on TD: ensemble disagreement, the absolute value level, the within-episode clock, and a state-visit count all keep moving long after the value function has settled. Familiarity is the most discriminative dimension in our experiments, and the reason is the same. It tracks how often the agent has been in a state, which is a strictly stronger signal than the instantaneous TD residue at that state.

B. The Anticipation/Power Coupling Is Honest, Not a Bug

One residual tension survives the decorrelation audit: the VIF on anticipation reaches 7.4. The cause is structural rather than formulaic. Once the dueling head has converged, $V(s)$, the action-value spread, and the across-head dispersion become coupled in any state where the world has effectively revealed the value—the agent is near a terminal, the actions are roughly equally good, and the heads agree. We keep anticipation for two reasons. Conceptually it measures something neither of the other two does: absolute prospect, not the action-effect spread or the epistemic uncertainty. And operationally, dropping it costs little—a 7-dimensional ablation still beats the baseline by a wide margin. We choose to report the high VIF rather than mask it: it is a property of the world the agent trained on, not an artefact of overlap in the formulas.

C. Why the RL Task Performance Is Unchanged

The appraisal extractor reads from the agent's internal state but does not feed back into the policy. With the seed and the hyperparameters held fixed, the two trained policies are therefore numerically identical; only the appraisal logging differs between runs. The isolation is by design—it

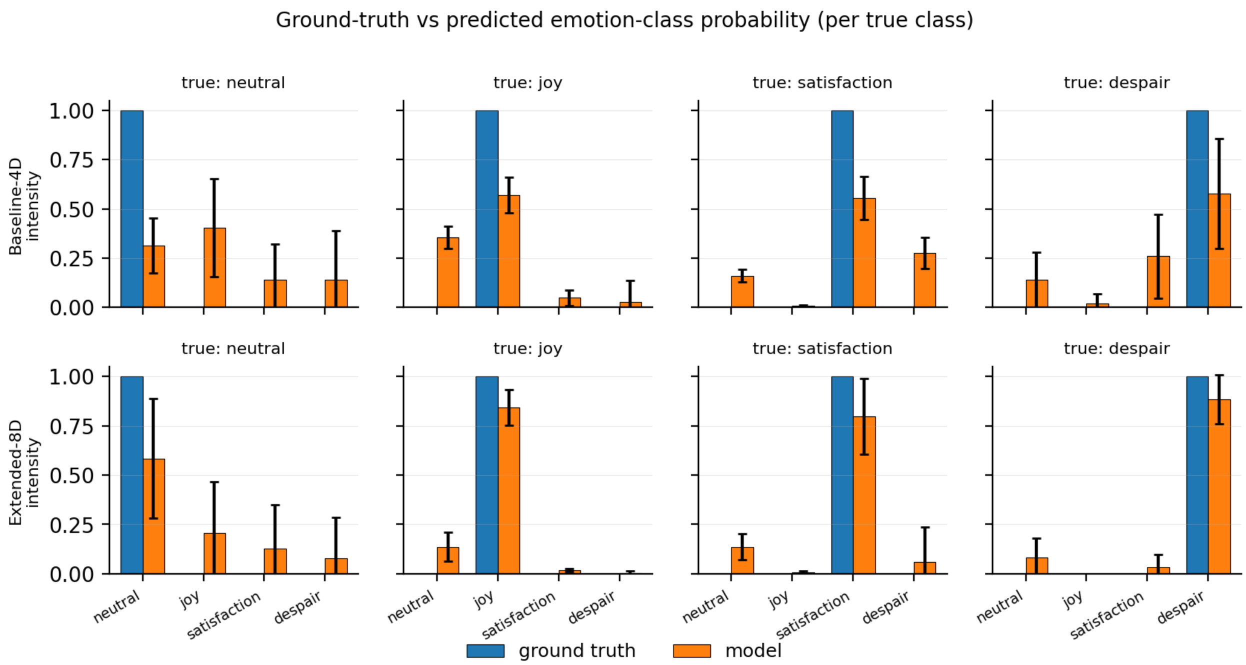

Fig. 8: Per-class probability calibration on the held-out test set, in the per-class reporting style of [15] (Figs. 4–7). Each cluster shows the model's mean predicted probability across the four emotion classes (orange, with ± one standard deviation whiskers) for held-out instances whose true label is given in the panel title; the ground-truth one-hot intensity is overlaid in blue for reference. The Extended-8D model produces a substantially sharper diagonal profile than the Baseline-4D model.

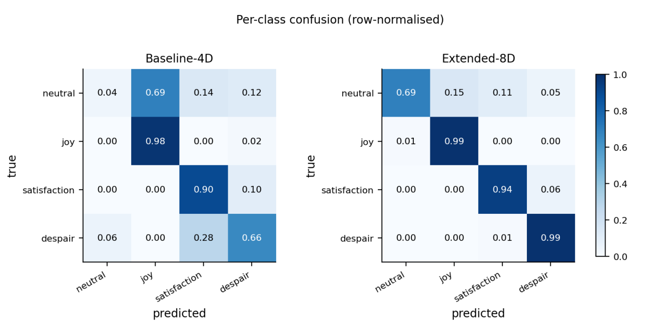

Fig. 9: Row-normalised confusion matrices on the held-out test set. The four-dimensional vector confounds neutral with joy and despair with satisfaction; the eight-dimensional vector resolves both confusions.

strips out any policy-side confound from the comparison—but it also means that the present formulation does not yet exploit appraisal as an exploration signal. Closing that loop is one of the items in Section XI.

D. Limitations

Three caveats deserve to be stated plainly. First, the emotion labels here are task-event proxies rather than human ratings. Aligning the setup with the vignette protocol of [15] would make the R^2 and RMSE numbers directly comparable with theirs. Second, the ensemble-disagreement signal that drives predictability is informative only after the heads have updated enough to disagree in a structured way; very early in training the dimension is close to constant and earns little. Third, the grid world and the four event classes were picked for analytical traceability. Whether the same eight dimensions retain their advantage on richer environments and a finer modal-emotion vocabulary is a question the present results do not directly answer; we treat it as future work.

X. Conclusion

We asked a narrow question and pursued it carefully: can the appraisal vector of a CPM-grounded RL agent be enriched with channels whose informativeness is empirically verifiable rather than asserted? Replacing the original tabular Q-learning with a Dueling Double DQN that carries a bootstrap ensemble lets us define four new channels—predictability, anticipation, urgency and familiarity—each from a statistic of the agent's experience that is algebraically disjoint from the original four. On a controlled grid-world benchmark, the resulting eight-dimensional vector roughly doubles the effective rank of the appraisal covariance, lifts classification accuracy from 64.8% to 90.3%, and improves R^2 from 0.366 to 0.757, all without changing the agent's task return. The gain traces back to the new dimensions: a permutation analysis flags four of them in the top five for discriminative use, which is the strongest direct evidence that the added capacity is informational rather than parametric. We also do not hide the one residual redundancy—a high VIF on anticipation under converged policies—and instead explain why it is structural rather than a modelling oversight.

XI. Future Scope

A handful of extensions follow from where the present work stops short. The most direct is to port the formulation onto the vignette protocol of [15], which would put the reported R^2 and RMSE on the same footing as their human-rating numbers. A natural second step is to train the appraisal classifier and the value network jointly, in place of the current read-out-then-classify pipeline; an end-to-end objective would let the appraisal subspace align with whatever discriminative geometry the classifier ends up using. A third would close the loop the original CPM hints at, feeding the appraisal vector back into the policy as an intrinsic-reward signal so that the agent learns from the very emotions it computes. A fourth concerns scaling: the count tables behind suddenness and familiarity become impractical on high-dimensional observations, so a hash-based pseudo-count or a successor-feature density is the obvious replacement. Finally, the remaining CPM checks—external causation, norm conformity, control attribution—can be operationalised on the same backbone whenever a clean source statistic for each can be identified; we leave that as incremental, not a rewrite.

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